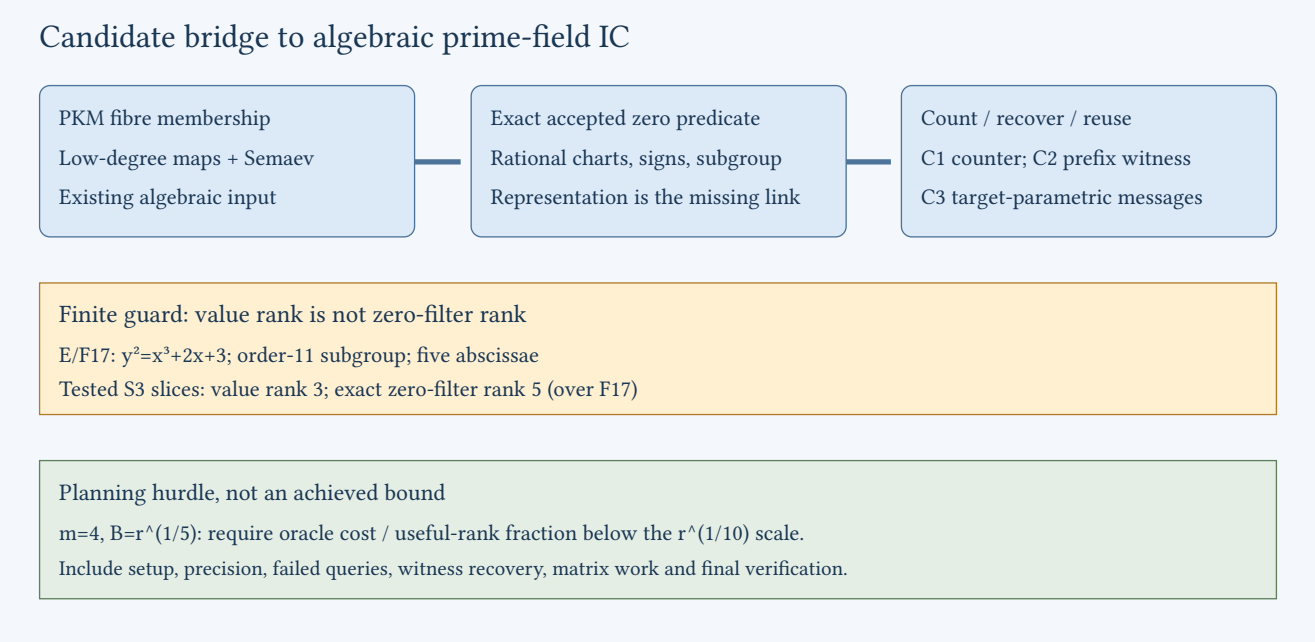
Candidate ideas for algebraic prime-field index calculus

2026-10-30 · proposals and exact finite controls; novelty not established

The proposed bridge combines PKM membership chains with weighted/exact counting on the actual rational, subgroup-valid zero predicate. A compact polynomial-value representation alone does not supply this predicate or an algorithm for locating its zeros. No new IC algorithm, speedup or proof is claimed. The preceding final-slot oracle remains a generic technique.



C1 — Exact predicate contraction. Feed PKM auxiliary-isogeny membership and summation into a bounded exact counter. Audit decoded Boolean size, coefficient algebra, precision and actual intermediate storage. Keep Kummer/Dickson controls and their existing leading-form obstruction. The integration is unimplemented.

C2 — Rational-chart-aware witnesses. Combine count-guided prefix extraction with a complete chart cover, rational lifts and subgroup membership. Avoid whole-grid recovery only after demonstrating complete agreement with direct small-field group addition, including poles and intermediate identity cases. Saturation alone can discard valid charts. This combination is unimplemented.

C3 — Target-parametric messages. Reuse exact target-independent membership messages across the relation queries of one solve. Existing coefficient specialization is prior work; a new compact message representation is needed. A table of all targets or tuples is not such a representation. Charge cold setup.

Finite controls. Nine native catalog/control tests and strict Clippy pass. For the tested order-11 subgroup over F17, S3 values have rank3 while their zero filters have rank5. These coordinate-field ranks are not useful relation rank over F11. Small slices do not prove asymptotic rank growth or rule out compact functional predicates. No performance experiment was run.

Cost hurdle. Under explicitly heuristic decomposition and useful-rank assumptions, the README includes setup, failed queries, witness recovery and quadratic matrix work. For the hypothetical m=4, B=r^(1/5), the oracle cost divided by its useful-rank fraction must beat the r^(1/10) scale. This is a requirement for a future solver, not a demonstrated rate.

Sources and gaps. PKM2016 sections3.1–3.4; Hexagon2610.00594v1, 2610.00613v1 and2610.00600v1; Kosters–Yeo1503.08001v3; existing prime-IC/tower studies; a public S3-value-rank README. Source proofs and external timings were not independently validated. A relevant2018 primary repository page returned403; its full text remains unreviewed. Search absence does not prove novelty.